

Modulus Functions & the Factor/Remainder Theorem

ANSWER KEY



Evaluating and solving modulus equations

- 1 Solve $|2x - 3| = 7$.
Either $2x - 3 = 7$, giving $x = 5$, or $2x - 3 = -7$, giving $x = -2$. Solutions: $x = 5$ or $x = -2$.
- 2 Solve $|x - 4| = |2x + 1|$.
Either $x - 4 = 2x + 1$, giving $x = -5$, or $x - 4 = -(2x + 1)$, i.e. $x - 4 = -2x - 1$, so $3x = 3$, $x = 1$.
Solutions: $x = -5$ or $x = 1$.

Modulus inequalities

- 3 Solve the inequality $|3x + 2| < 8$, giving your answer as an interval.
 $-8 < 3x + 2 < 8$, so $-10 < 3x < 6$, giving $-\frac{10}{3} < x < 2$.
- 4 Solve $|x - 5| \geq 3$, giving your answer as a union of intervals.
Either $x - 5 \geq 3$ (giving $x \geq 8$) or $x - 5 \leq -3$ (giving $x \leq 2$). Solution: $x \leq 2$ or $x \geq 8$.

Visualizing the graph of $y = |f(x)|$

- 5 The graph shows $y = |x^2 - 4|$.
 - a Explain how this graph relates to the graph of $y = x^2 - 4$, and identify which portion has been reflected. $y = x^2 - 4$ is a parabola with vertex $(0, -4)$, dipping below the x -axis between $x = -2$ and $x = 2$. Taking the modulus reflects the portion of the graph that was below the x -axis (between $x = -2$ and $x = 2$) upward, while the portions already above the axis (for $|x| > 2$) stay unchanged.
 - b State the x -values where the graph touches the x -axis, and explain why the curve has sharp corners there rather than smooth turning points. The graph touches the x -axis at $x = -2$ and $x = 2$, where the original function $x^2 - 4$ equals zero. Sharp corners (not smooth turning points) occur there because the modulus function abruptly changes from following $-(x^2 - 4)$ to following $x^2 - 4$ at those points, creating a kink in the graph.

The factor theorem

- 6 Use the factor theorem to show that $(x - 2)$ is a factor of $f(x) = x^3 - 3x^2 - 4x + 12$, then factor $f(x)$ completely.
 $f(2) = 8 - 12 - 8 + 12 = 0$, confirming $(x - 2)$ is a factor. Dividing:
 $x^3 - 3x^2 - 4x + 12 = (x - 2)(x^2 - x - 6) = (x - 2)(x - 3)(x + 2)$.

- 7 Find the value of k such that $(x + 3)$ is a factor of $f(x) = x^3 + kx^2 - 5x - 6$.
By the factor theorem, $f(-3) = 0$: $(-3)^3 + k(-3)^2 - 5(-3) - 6 = -27 + 9k + 15 - 6 = 9k - 18 = 0$, giving $k = 2$.
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The remainder theorem

- 8 Find the remainder when $f(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $(x - 2)$.
By the remainder theorem, the remainder equals $f(2)$:
 $f(2) = 2(8) - 5(4) + 3(2) - 7 = 16 - 20 + 6 - 7 = -5$.
- 9 When $f(x) = x^3 + ax^2 + bx - 4$ is divided by $(x - 1)$ the remainder is 2, and when divided by $(x + 2)$ the remainder is -22 . Find a and b .
 $f(1) = 1 + a + b - 4 = a + b - 3 = 2$, so $a + b = 5$.
 $f(-2) = -8 + 4a - 2b - 4 = 4a - 2b - 12 = -22$, so $4a - 2b = -10$, i.e. $2a - b = -5$. Adding to $a + b = 5$: $3a = 0$, so $a = 0$, then $b = 5$.
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Solving equations involving modulus functions graphically

- 10 Solve $|x - 3| = 2x + 1$, checking that each solution satisfies the original equation (since squaring or case-splitting can introduce extraneous solutions).
Case 1 ($x \geq 3$): $x - 3 = 2x + 1$, giving $x = -4$ — but this contradicts $x \geq 3$, so reject. Case 2 ($x < 3$):
 $-(x - 3) = 2x + 1$, so $3 - x = 2x + 1$, giving $3x = 2$, $x = \frac{2}{3}$. Check: $\left|\frac{2}{3} - 3\right| = \frac{7}{3}$ and $2\left(\frac{2}{3}\right) + 1 = \frac{7}{3}$ — matches. Solution: $x = \frac{2}{3}$.
- 11 Solve the inequality $|2x + 3| \leq |x - 1|$ by squaring both sides (valid since both sides are non-negative).
Squaring: $(2x + 3)^2 \leq (x - 1)^2$. Expand: $4x^2 + 12x + 9 \leq x^2 - 2x + 1$, so $3x^2 + 14x + 8 \leq 0$. Factor: $(3x + 2)(x + 4) \leq 0$. This holds for $-4 \leq x \leq -\frac{2}{3}$.
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Finding all zeros of a higher-degree polynomial

- 12 Given that $x = 1$ and $x = -1$ are both zeros of $f(x) = x^4 - 5x^3 + 5x^2 + 5x - 6$, find the remaining two zeros.
Since $x = 1$ and $x = -1$ are zeros, $(x - 1)(x + 1) = x^2 - 1$ is a factor. Dividing $f(x)$ by $x^2 - 1$ gives $x^2 - 5x + 6 = (x - 2)(x - 3)$. So all four zeros are $x = 1, -1, 2, 3$; the remaining two are $x = 2$ and $x = 3$.
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Using the remainder theorem to test multiple divisors at once

- 13** A polynomial $f(x)$ leaves remainder 5 when divided by $(x - 2)$ and remainder -3 when divided by $(x + 1)$. Find the remainder when $f(x)$ is divided by $(x - 2)(x + 1)$, given that the remainder must be of the form $ax + b$.

The remainder $ax + b$ must satisfy $f(2) = 2a + b = 5$ and $f(-1) = -a + b = -3$. Subtracting: $3a = 8$, so $a = \frac{8}{3}$. Then $b = 5 - 2a = 5 - \frac{16}{3} = -\frac{1}{3}$. Remainder: $\frac{8}{3}x - \frac{1}{3}$.

Solving polynomial equations with the factor theorem

- 14** Solve $2x^3 - x^2 - 13x - 6 = 0$ completely, given that $x = -2$ is one root.

Since $x = -2$ is a root, $(x + 2)$ is a factor. Dividing: $2x^3 - x^2 - 13x - 6 \div (x + 2) = 2x^2 - 5x - 3$.

Factor the quadratic: $2x^2 - 5x - 3 = (2x + 1)(x - 3)$. So the equation factors as

$(x + 2)(2x + 1)(x - 3) = 0$, giving roots $x = -2$, $x = -\frac{1}{2}$, $x = 3$.

Graphing modulus functions with restrictions

- 15** Sketch, in words, the key features of $y = |2x - 4| + 1$: state its minimum value, the x -value where it occurs, and its y -intercept.

The minimum value of $|2x - 4|$ is 0, occurring at $x = 2$ (where $2x - 4 = 0$), so the minimum of $y = |2x - 4| + 1$ is 1, occurring at $x = 2$. The graph forms a V-shape with vertex $(2, 1)$. The y -intercept: at $x = 0$, $y = |-4| + 1 = 5$.

Finding unknown coefficients using both theorems together

- 16** The polynomial $f(x) = x^3 + ax^2 + bx + 6$ has $(x - 1)$ as a factor and leaves a remainder of -12 when divided by $(x + 2)$. Find a and b .

By the factor theorem, $f(1) = 0$: $1 + a + b + 6 = 0$, so $a + b = -7$. By the remainder theorem, $f(-2) = -12$: $-8 + 4a - 2b + 6 = -12$, so $4a - 2b = -10$, i.e. $2a - b = -5$. Adding this to $a + b = -7$: $3a = -12$, so $a = -4$; then $b = -7 - (-4) = -3$.

- 17** Show that $f(x) = x^3 - 6x^2 + 11x - 6$ has three distinct real roots by testing $x = 1, 2, 3$ using the factor theorem, and write $f(x)$ in fully factored form.

$f(1) = 1 - 6 + 11 - 6 = 0$ ✓. $f(2) = 8 - 24 + 22 - 6 = 0$ ✓. $f(3) = 27 - 54 + 33 - 6 = 0$ ✓. Since $f(x)$ is a cubic (degree 3) and has three distinct roots $x = 1, 2, 3$, it is completely factored as $f(x) = (x - 1)(x - 2)(x - 3)$.